Midterm Exam

Introduction to Statistics (통계의 이해) **2026 2nd Semester**

Year(학년): ___ Section(교반): ___ Cadet Number(교번): _____ Name(성명): _____ Score: _____

- **All solutions must include a detailed step-by-step explanation.**
- **If an answer has five or more decimal places, round to the fourth decimal place.**
- **Use the reference materials provided on the last page of the exam.**

1. Write O if each statement is true and X if it is false. [7 points]

(1) Adding 5 to every observation increases the mean by 5 but does not change the variance. ( )

(2) A probability density function $f(x)$ always satisfies $0 \leq f(x) \leq 1$. ( )

(3) If $f(x, y)$ is the joint probability mass function of X and Y, then the marginal probability mass function of X is $f_X(x) = \sum_{\forall y} f(x, y)$. ( )

(4) If $X \sim B(1, p)$, then $E(X) = p$ and $\mathrm{Var}(X) = p(1-p)$. ( )

(5) If $X \sim N(\mu, \sigma^2)$, then $\dfrac{X - \mu}{\sigma} \sim N(0, 1)$. ( )

(6) By the Central Limit Theorem, the population distribution becomes approximately normal as the sample size increases. ( )

(7) An unbiased estimator gives an estimate equal to the true value of the parameter. ( )

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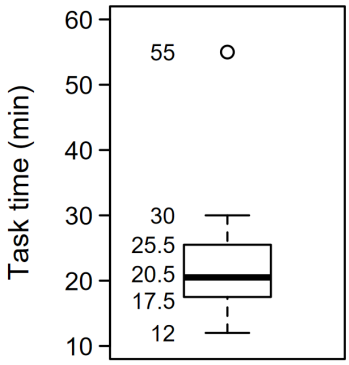
2. The plots below show the task times (in minutes) of 20 cadets. Fill in each blank or choose the correct answer in the parentheses. [10 points]

Task times of 20 cadets

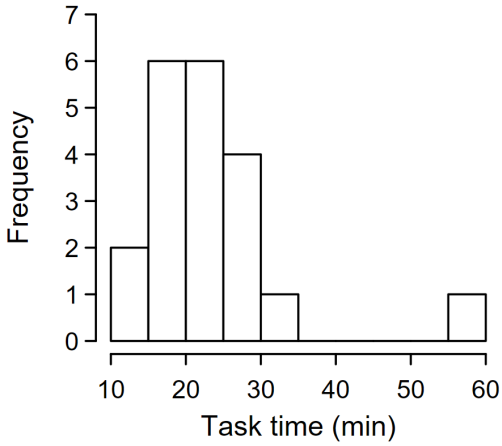
Stem-and-leaf plot

Stem	Leaf
1	2 4 5 6 7 8 8 9
2	0 0 1 2 3 4 5 6 7 8
3	0
4	
5	5

Box plot



Histogram



(Stem-and-leaf key: 1 | 2 means 12 minutes.)

(1) The number of cadets who took exactly 20 minutes is ( ).

(2) The median is ( ) minutes.

(3) The interquartile range (IQR) is ( ) minutes.

(4) If the bin width changes from 5 to 10 minutes, the histogram (can look different / must look the same), and the mean of the data (changes / does not change).

(5) The outlier 55 was a recording mistake, and the correct value is 40. After this correction, the median (increases / does not change / decreases), and the mean (increases / does not change / decreases).

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3. The probability that an aircraft is in a radar's watch area is 0.2. If an aircraft is in the area, the radar sounds an alarm with probability 0.9. If not, it still sounds an alarm with probability 0.15. Answer the following questions. [10 points]

(1) Find the probability that the radar sounds an alarm. (Define the events first.)

(2) If the radar sounds an alarm, find the probability that an aircraft is really in the area.

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4. The joint probability density function of X and Y is

$$f(x,y) = \begin{cases} c\,(1+x^2 y), & -1 < x < 1,\ -1 < y < 1, \\ 0, & \text{otherwise.} \end{cases}$$

Answer the following questions. [15 points]

(1) Find the constant c.

(2) Find the marginal probability density functions of X and Y.

(3) Find $E(X)$, $E(Y)$, and $E(XY)$.

(4) Find $\mathrm{Cov}(X,Y)$.

(5) Determine whether X and Y are independent.

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5. The weights (in grams) of products from Company A and Company B follow normal distributions with means $\mu_A = 500$, $\mu_B = 480$ and standard deviations $\sigma_A = 20$, $\sigma_B = 15$. One product is chosen independently from each company. Find the probability that the Company A product is at least 45 g heavier than the Company B product. (Define the random variables first.) [8 points]

6. The scores on a test follow a normal distribution with mean $\mu = 72$ points and standard deviation $\sigma = 9$ points. The top 5% of students get an A. Find the cutoff score for an A. (Define the random variable first.) [8 points]

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7. The lifetime X (in 1,000 hours) of a military radar part has the probability density function

$$f(x) = \begin{cases} \frac{1}{2}e^{-x/2}, & x > 0, \\ 0, & \text{otherwise.} \end{cases}$$

Answer the following questions. [15 points]

(1) Show that $E(X) = 2$. (Note: $E(X^2) = 8$.)

(2) For a random sample $X_1, \ldots, X_{64}$ of $n = 64$ parts, find the approximate distribution of the sample mean $\bar{X}$, and name the theorem you used.

(3) Find the approximate value of $P(1.5 \leq \bar{X} \leq 2.5)$.

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8. The number of defects X on a piece of military equipment has the following probability distribution.

x	0	1	2
$P(X=x)$	5/8	2/8	1/8

Let $\bar{X}_n$ be the sample mean of a random sample $X_1, \ldots, X_n$. Answer the following questions. [12 points]

(1) For $n = 2$, find the probability distribution of $\bar{X}_2$. Then find $P(\bar{X}_2 \geq 0.8)$.

(2) For $n = 50$, use the Central Limit Theorem to find the approximate distribution of $\bar{X}_{50}$. Then find the approximate value of $P(\bar{X}_{50} \geq 0.8)$.

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9. Let $X_1, X_2, \ldots, X_n$ be a random sample from the uniform distribution on the interval $(\theta,\ \theta + 1)$. Answer the following questions. [15 points]

(1) Find $E(X_1)$ and $\mathrm{Var}(X_1)$.

(2) Find $E(\bar{X})$ and $\mathrm{Var}(\bar{X})$, where $\bar{X}$ is the sample mean.

(3) Using $\bar{X}$, find an estimator $\hat{\theta}$ of θ that is unbiased and consistent. Show both properties in detail.

※ **Reference Table** Let $Z \sim N(0,1)$ and $\Phi(z) = P(Z \le z)$.

$\Phi(0.25) = 0.5987$	$\Phi(0.5) = 0.6915$	$\Phi(0.75) = 0.7734$	$\Phi(1) = 0.8413$
$\Phi(1.25) = 0.8944$	$\Phi(1.5) = 0.9332$	$\Phi(1.75) = 0.9599$	$\Phi(2) = 0.9772$
$\Phi(2.25) = 0.9878$	$\Phi(2.5) = 0.9938$	$\Phi(2.75) = 0.9970$	$\Phi(3) = 0.9987$
$z_{0.30} = 0.5244$	$z_{0.20} = 0.8416$	$z_{0.15} = 1.0364$	$z_{0.10} = 1.2816$
$z_{0.05} = 1.6449$	$z_{0.025} = 1.9600$	$z_{0.01} = 2.3263$	$z_{0.005} = 2.5758$
$\chi^2_{0.05}(7) = 14.067$	$\chi^2_{0.025}(7) = 16.013$	$\chi^2_{0.05}(8) = 15.507$	$\chi^2_{0.025}(8) = 17.535$
$\chi^2_{0.05}(9) = 16.919$	$\chi^2_{0.025}(9) = 19.023$	$\chi^2_{0.05}(10) = 18.307$	$\chi^2_{0.025}(10) = 20.483$